Deformation Theory of Calabi-Yau Varieties

Undergraduate Thesis Proposal

Weining Huang

Beihang University
School of Mathematical Sciences

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Outline

1 Preliminaries

2 Smooth Calabi-Yau Manifolds

3 Threefold Calabi-Yau Varities with ODP

4 Expected Outcomes and References

Convention

- Usually 'Variety' means a seperated integral scheme of finite type over a fixed field k. Here we can think $k = \mathbb{C}$.
- But for convenience we assume all varieties appear in the slides are projective.
- Thus we can regard varieties over $\mathbb{C}$ as complex compact Kähler manifolds and use tools in both algebraic geometry and complex geometry to investigate by GAGA.

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- First order deformations of complex structure (of X) are described by **Kodaira-Spencer class** in $H^1(T_X)$.
- Deformations are governed by the **Maurer-Cartan equation**:

$$\bar{\partial}\varphi + \frac{1}{2}[\varphi, \varphi] = 0, \quad \varphi \in \mathcal{A}^{0,1}(T_X)$$

Deformation Theory with Algebraic Approach

Let X be a scheme over a fixed field $k(=\mathbb{C})$.

- Define the deformation functor of X:

$$\mathrm{Def}_X := \mathrm{C} \to \mathrm{Set},\ A \mapsto \{\text{deformations of } X \text{ over } A\}.$$

Here C is he category of local artinian k-algebras with residue field k.

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- **Schlessinger's Criterion** helps to investigate properties of deformation functor.

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- Complex-geometric proof: Calabi-Yau theorem + Tian-Todorov lemma.

Algebraic proof for BTT Theorem

Theorem (Z. Ran, Y. Kawamata, 1992)

Let X be a scheme over a fixed field $k(=\mathbb{C})$. If the deformation functor Def_X satisfies the T^1 lifting property, then deformations of X are unobstructed.

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- It can be proved that for X being a nonsingular Calabi-Yau variety, Def_X satisfies the T^1 lifting property.
- Ziv Ran developed the purely algebraic approach, and Yujiro Kawamata modified the T^1 lifting property to more natural version.

Deformations of Threefold CY varieties with ODP

Let's shift our scope to singular and threefold case. Usually our main aim is to deform a singular variety to a nonsingular one.

Definition (Original Double Point)

An **ODP** is an isolated singularity in a threefold locally analytically isomorphic to the origin of

$$\{ (z_1, \ldots, z_4) \in \mathbb{C}^4 \mid z_1^2 + z_2^2 + \cdots + z_4^2 = 0 \}.$$

Deformations of Threefold CY varieties with ODP

Let X_0 be a threefold Calabi-Yau variety in the sense that dualizing sheaf $\omega_{X_0} \cong \mathcal{O}_{X_0}$, with only ODP as singularities. Let $X \to X_0$ be a small resolution, with exceptional curves $C_1, \ldots, C_r$, and Let $[C_i]$ be the class of C_i in $H^4(X, \mathbb{C})$.

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Theorem (Friedman, 1986)

X_0 is smoothable if

- $[C_1], [C_r]$ *generate* $H^2(X, \Omega_X^2)$
- *There exists a relation* $a_1[C_1] + \cdots + a_r[C_r] = 0 \in H^4(X, \mathbb{C})$ *with* $a_i \neq 0$ *for every* i.

X_0 is smoothable means X_0 can be deformed to a nonsingular variety X_t.

Deformations of Threefold CY varieties with ODP

- Subsequently, independent work of Y. Kawamata, Z. Ran and G. Tian showed that the first assumption is not necessary provided that there exists a resolution $X' \to X_0$ where X' is Kähler.
- The key point in removing this assumption is a generalization of the unobstructedness theorem for smooth Calabi-Yau manifolds, in any dimension:

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Theorem (Kawamata, Ran, Tian)

Let X_0 be a Calabi-Yau variety in the sense that dualizing sheaf $\omega_{X_0} \cong \mathcal{O}_{X_0}$, with only ODP as singularities and there exists a resolution $X' \to X_0$ where X' is Kähler. Then deformations of X_0 are unobstructed.

Expected Outcomes

- Understand the basic framework of deformation theory.

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- Understand the basic framework of deformation theory.
- Master the key results in deformation theory of both smooth and singular Calabi-Yau varieties.

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Thank you!

Questions and comments are welcome.